MA 102
Ordinary Differential Equations
Lecture 17
(April 16, 2024)

By studying a linear system, we can learn a lot about various properties of a nonlinear system. Although our aim does not lie in studying nonlinear systems, we need to learn a few aspects of nonlinear systems which will help us in realizing or understanding a linear system better.

Examples of Nonlinear Equations

1. If θ is the angle of deviation of an undamped pendulum of length a whose bob has mass m, the resulting equation of motion is represented by

$$\frac{d^2\theta}{dt^2} + \frac{g}{a}\sin\theta = 0. \tag{1}$$

2. If a damping force proportional to the velocity of the bob is present, then (1) becomes

$$\frac{d^2\theta}{dt^2} + \frac{c}{a}\frac{d\theta}{dt} + \frac{g}{a}\sin\theta = 0. \tag{2}$$

   The above two equations can be linearized by taking $\sin\theta$ to be θ, which is reasonable for small oscillations but is highly unreasonable for large deflections.

3. The theory of vacuum tube leads to van der Pol equation:

$$\frac{d^2x}{dt^2} + \mu(x^2-1)\frac{dx}{dt} + x = 0. \tag{3}$$

All of equations (1), (2) and (3) are second-order nonlinear equations which are of the form

$$\frac{d^2x}{dt^2} = f\left(x, \frac{dx}{dt}\right). \tag{4}$$

If we imagine a simple dynamical system consisting of a particle of unit mass moving on the x-axis, and if $f\left(x, \frac{dx}{dt}\right)$ is the force acting on it, then (4) is the equation of motion of the particle. The values of x (position) and $\frac{dx}{dt}$ (velocity), which at each instant characterize the state of the system, are called its phases, and the plane of the variables x and $\frac{dx}{dt}$ is called the phase plane.

If we introduce the variable $y = dx/dt$, then (4) can be replaced by the equivalent first-order system

$$\left.\begin{aligned} \tfrac{dx}{dt} &= y, \\ \tfrac{dy}{dt} &= f(x, y). \end{aligned}\right\} \tag{5}$$

We can learn a good deal about the solutions of (4) by studying the solutions of (5). When t is regarded as a parameter, then in general a solution of (5) is a pair of functions $x(t)$ and $y(t)$ defining a curve in the xy-plane, which is simply the phase plane.

More generally, we study systems of the form

$$\left.\begin{aligned} \frac{dx}{dt} &= F(x,y), \\ \frac{dy}{dt} &= G(x,y), \end{aligned}\right\} \tag{6}$$

where F and G are continuous and have continuous first partial derivatives throughout the plane.

A system like this in which the independent variable t does not appear in the functions F and G is said to be autonomous.

We know that if t_0 is any number and (x_0, y_0) is any point in the phase plane, then there exists a unique solution

$$x = x(t), \ y = y(t) \tag{7}$$

of (6) such that $x(t_0) = x_0, \ y(t_0) = y_0$. If $x(t)$ and $y(t)$ both are not constant functions, then (7) defines a curve in the phase plane called a path or trajectory of the system.

In general, the paths of (6) cover the entire plane and do not intersect one another. The only exception occurs at points (x_0, y_0) where both F and G vanish, i.e., when

$$F(x_0, y_0) = 0, \quad G(x_0, y_0) = 0.$$

In other words,

both $\dfrac{dx}{dt}$ and $\dfrac{dy}{dt}$ are zero at such points (x_0, y_0).

These points are called critical points, and at such a point, the unique solution is the constant solution $x = x_0$ and $y = y_0$. A constant solution does not define a path and therefore, no path goes through a critical point.

We shall assume that each critical point (x_0, y_0) is isolated in the sense that there exists a small circle centred on (x_0, y_0) that contains no other critical point.

A point $P_0(x_0, y_0)$ is called a stable critical point of (6) if all paths of (6) which at some instant are sufficiently close to P_0 remain close to P_0 at all future times.

More precisely,

if for every disk D_ϵ of radius $\epsilon > 0$ about P_0, there is a disk D_δ of radius $\delta > 0$ about P_0 such that every path of (6) which has a point P_1 (corresponding to $t = t_1$, say) in D_δ has all its points corresponding to $t \geq t_1$ in D_ϵ. (Refer to Figure 1 on next page.)

Stability of linear systems

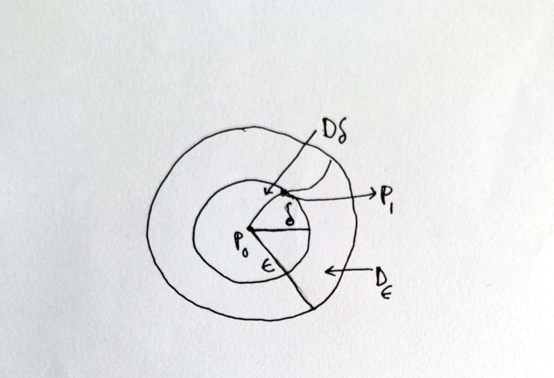

Figure 1

P_0 is called a stable and attractive critical point of (6) if P_0 is stable and every path which has a point in D_δ approaches P_0 as $t \to \infty$.
P_0 is called unstable if P_0 is not stable.

To obtain a physical interpretation of critical points, let us consider the special autonomous system (5) arising from the dynamical system (4). In this case, a critical point is a point $(x_0, 0)$ at which $y = 0$ and $f(x_0, 0) = 0$.

That is,

the critical point corresponds to that state of the motion of the particle in which both the velocity dx/dt and the acceleration $dy/dt = d^2x/dt^2$ vanish. This means that the particle is at rest with no force acting on it, and is therefore in a state of equilibrium.

It is evident that the states of equilibrium of a physical system are the most important features and they lead to our interest in studying critical points.

The general autonomous system (6) does not necessarily arise from any dynamical equation of the form (4). Now let us consider a vector field defined by

$$\mathbf{V}(x,y) = F(x,y)\hat{\mathbf{i}} + G(x,y)\hat{\mathbf{j}},$$

which at a typical point $P = (x,y)$ has horizontal component $F(x,y)$ and vertical component $G(x,y)$.

Since $dx/dt = F$ and $dy/dt = G$, this vector is tangent to the path at P and points in the direction of increasing t. If we think of t as time, then $\mathbf{V}$ can be interpreted as the velocity vector of a particle moving along the path.

We can also imagine that the entire phase plane is filled with particles, and that each path is the trail of a moving particle preceded and followed by many others on the same path and accompanied by yet others on nearby paths. The situation can be described as a two-dimensional fluid motion; and since the system (6) is autonomous, which means that the vector $\mathbf{V}(x,y)$ at a fixed point (x,y) does not change with time, the fluid motion is stationary.

The paths are the trajectories of the moving particles, and the critical points Q, R and S are points of zero velocity where the particles are at rest, (i.e., the stagnation points of the fluid motion).

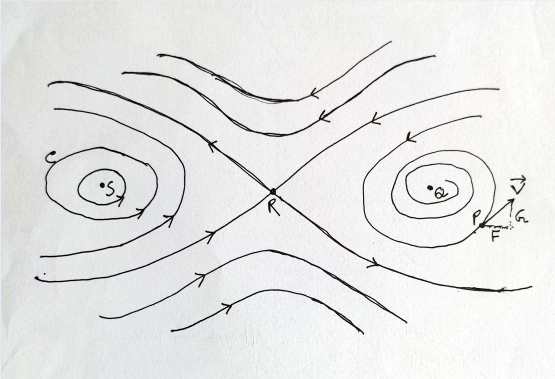

Figure 2

The most striking features of the fluid motion can be observed from Figure 2 as

1. the existence of critical points,
2. the arrangement of the paths near critical points,
3. the stability or instability of the critical points, that is, whether a particle near such a point remains near or goes away to another part of the plane,
4. closed paths like C, which correspond to periodic solutions.

Such features constitute a major part of the phase portrait of the system (6). In general, nonlinear equations and systems cannot be solved explicitly and hence the objective here is to find as much information as possible about the phase portrait directly from the functions F and G.

We can observe that if $x(t)$ is a periodic solution of the dynamical equation (4), then its derivative $y(t) = dx/dt$ is also periodic and the corresponding path of the system (5) is therefore closed.

Conversely, if any path of (5) is closed, then (4) has a periodic solution. For example, van der Pol equation, although it cannot be solved, can be shown to have a unique periodic solution, for $\mu > 0$, by showing that its equivalent autonomous system has a unique closed path.

Types of critical points

Consider the following autonomous system:

$$\left.\begin{aligned} \frac{dx}{dt} &= F(x,y), \\ \frac{dy}{dt} &= G(x,y), \end{aligned}\right\} \tag{8}$$

where the functions F and G are continuous and have continuous first-order partial derivatives throughout the xy-plane. The critical points can be found by solving $F(x,y)=0$ and $G(x,y)=0$. There are four simple types of critical points. But before discussing the critical points, we need to know certain other aspects as follows.

Definition: Let (x_0, y_0) be an isolated critical point of (8). If $\Gamma_c \equiv [x(t), y(t)]$ is a path of (8), then we say Γ_c approaches (x_0, y_0) as $t \to \infty$ if

$$\lim_{t\to\infty} x(t) = x_0, \quad \lim_{t\to\infty} y(t) = y_0. \tag{9}$$

Geometrically, this means that if $P = (x,y)$ is a point that traces out Γ_c in accordance with the equations $x = x(t)$ and $y = y(t)$, then $P \to (x_0, y_0)$ as $t \to \infty$.

For some cases, it is possible to find the explicit solutions of the system (8) and these solutions can be used to determine the paths. However, in most cases, in order to find the paths, it is necessary to eliminate t between the two constituent equations of the system to get

$$\frac{dy}{dx} = \frac{G(x, y)}{F(x, y)}. \tag{10}$$

This first-order equation (10) gives the slope of the tangent to the path of (8) that passes through the point (x, y) provided that the functions $F(x, y)$ and $G(x, y)$ are not both zero at this point.

In this case, of course, this point is a critical point and no path passes through it. The paths of (8) therefore coincide with the one-parameter family of orthogonal curves of (10) and this family can be obtained without much difficulty. However, it may be noted that while the paths of (8) are directed curves, the integral curves of (10) have no direction associated with them.

We will next discuss different types of critical points mainly with respect to geometrical interpretation. In most of the cases, the origin $O = (0, 0)$ is the point which is considered a critical point.